Flashing a New Image to the UNO Q

Learn how to flash a new image (the Linux OS) to the UNO Q board using the Arduino Flasher CLI.

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The Arduino® UNO Q runs a Linux operating system (OS), which comes pre-installed on the board.

There is often no need to re-install the OS on your board (also known as flashing an image), as the OS automatically receives updates regularly.

However, if we want to hard-reset the board and perform a fresh installation, it is possible. This process requires some additional tools installed on your machine.

> *WARNING! Note that the instructions in this tutorial will wipe the board clean, and files & configurations saved on the board will be destroyed.*

Requirements

Help

Hardware Requirements

Female-to-female jumper wire

Software Requirements

To re-flash the board, we will use the **Arduino Flasher CLI** tool:

- Download the **Arduino Flasher CLI**

Note that this tool will download an image with a size that exceeds 1 GB. A stable Internet connection is recommended.

Download & Install CLI Tool

1. Download the **Arduino Flasher CLI** for your OS (MacOS / Linux / Windows)
2. Unzip the downloaded file, (you will receive a executable binary named `arduino-flasher-cli`)

Verify Tool Is Installed

Before flashing a new image, check that the `arduino-flasher-cli` tool is working. Below are instructions to verify the tool is installed and working on your OS.

MacOS

Navigate to the unzipped folder

`arduino-flasher-cli-x.x.x-darwin-arm64`), and run the following command:

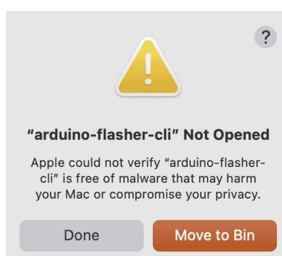
```
./arduino-flasher-cli
```

You should see something like:

Output from testing tool (MacOS)

This means it is working, and we can proceed to **flashing the board**.

Important Note: Do not run the file directly from Finder, you will receive a prompt window akin to:



Error running

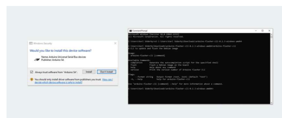
As the tool is ran from the command line with specific flags (explained further below), there is no reason to run it from Finder.

Windows

Navigate to the unzipped folder (e.g. `arduino-flasher-cli-x.x.x-darwin-arm64`

```
1  arduino-flasher-cli
```

A new window should appear, prompting you to install the driver. Install it, and run `arduino-flasher-cli` again in the terminal.



Output from testing tool (Windows)

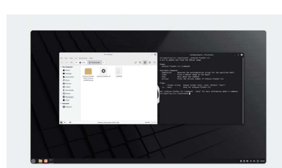
This means it is working, and we can proceed to **flashing the board**.

Linux

Navigate to the unzipped folder (e.g. `arduino-flasher-cli-x.x.x-darwin-arm64`), and run the following command:

```
1  ./arduino-flasher-cli
```

You should see something like:



Output from testing tool (Linux)

This means it is working, and we

Note: in some Linux systems, the `arduino-flasher-cli` could exit with an error right before flashing. This may occur if the kernel module `qcserial` is loaded. A workaround solution to fix this is in place (see section below).

Fixing `qcserial` Issue (Linux Only)

Open a terminal, and check if `qcserial` is present by running `lsmod | grep qcse`

If present, it will return:

```
> lsmod | grep qcse
qcserial
usb_wwan
usbserial
```

Then, check if `qcserial` is locking the serial port `ttyUSB0`, by running `sudo dmesg`:

```
> sudo dmesg
[31633.372270] qcserial
[31633.372308] qcserial
```

This issue can be fixed by disabling (or blacklisting), the `qcserial` kernel module on the Linux system.

1. Create or edit a

`blacklist-modem.conf` inside the `/etc/modprobe.d/` directory:

```
sudo nano /etc/
```

2. Inside this file, we need to blacklist the `qcserial` module:

```
blacklist qcser
```

3. Save the file and restart the system for the configuration to take effect.

Preparing the Hardware

To prepare the hardware for flashing, follow the instructions below:

1. Disconnect the board from your computer.
2. Add the female-to-female jumper cable between the two pins specified in the image:

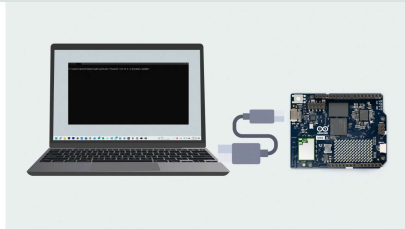
Short the two
pins

Flash Image to the Board

In this step, we will upload the new image to the board using the Arduino Flasher CLI.

1. Connect the board to your computer, using a USB-C® type cable.
2. Open a terminal and navigate to the directory where you unzipped the Arduino Flasher CLI (normally `cd /Downloads`).
3. Run the following command in the terminal: `./arduino-flasher-cli flash latest` .
4. A download sequence will begin (the image is >1 GB). Once the download is complete, it will flash the board with the new image. **Please note:** this will take several minutes. Do **not** disconnect the USB cable during this process.
5. Once flashing completes and the tool reports success, **power-cycle** the board (unplug and re-plug USB) so it boots the new OS.

The steps above are summarized in the graphic below:



Troubleshooting

- Ensure the Arduino Flasher CLI is unzipped and accessible.
- Check that the correct pins are shorted, and that they are shorted **before** connecting the board to the computer.
- Make sure the board has not finished booting when running the flashing command (`./arduino-flasher-cli flash latest`).
- Verify that you are running the command from the correct directory where the Arduino Flasher CLI was unzipped.
- After the tool reports a successful installation, **power-cycle** the board **with the jumper removed.**

Suggest changes

The content on docs.arduino.cc is facilitated through a public GitHub repository. If you see anything

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